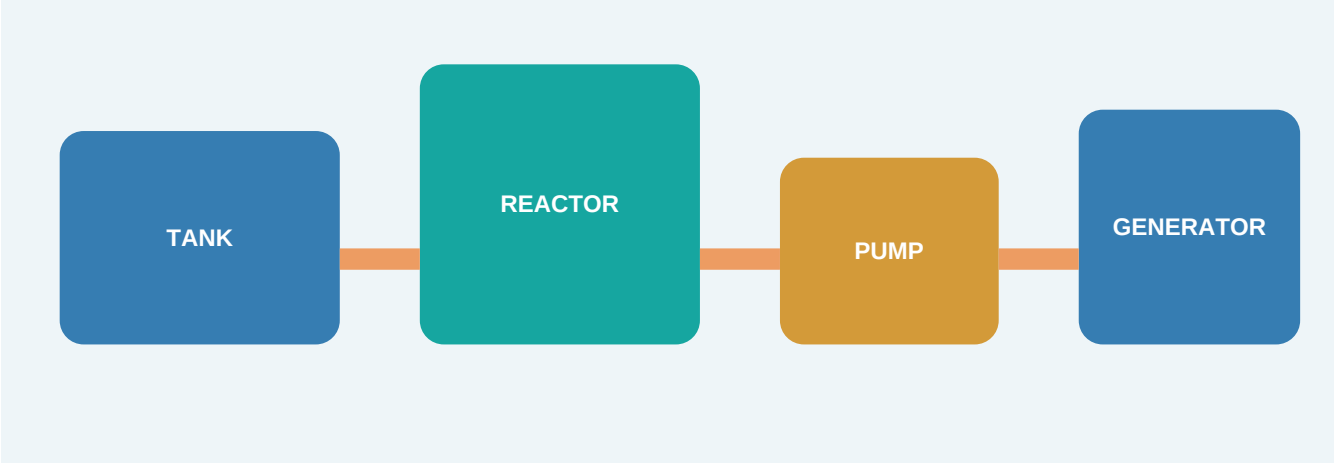
Planet Game - AssetForge Architecture

Version 1.0 | 29 October 2026 | Technical design proposal

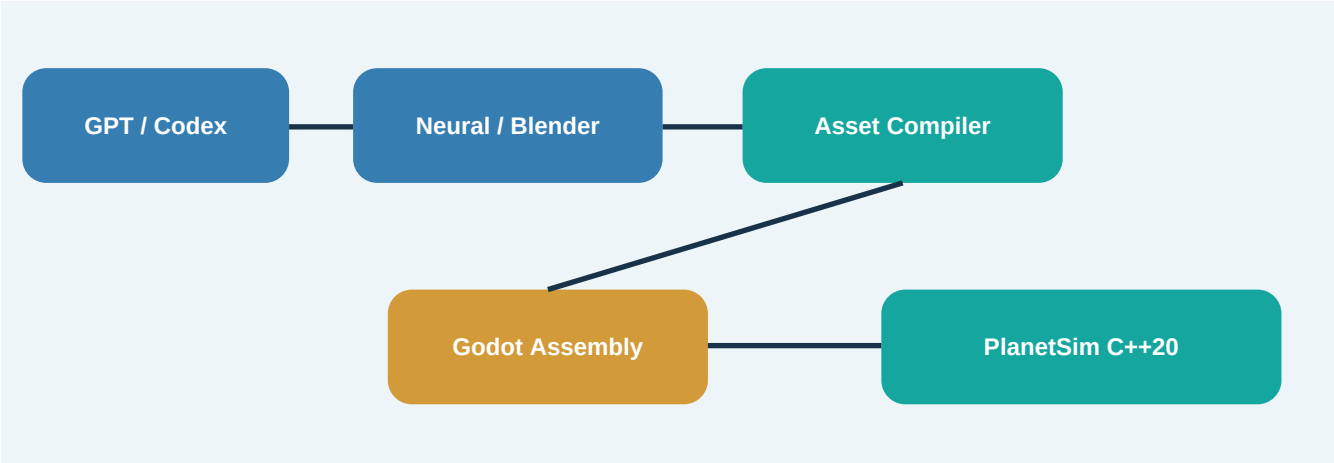


Executive summary

AssetForge is an AI-assisted *offline development* pipeline for Planet Game. GPT/Codex writes asset specifications and procedural Blender generators. Neural image-to-3D backends create irregular visual meshes. Blender normalizes, validates, optimizes and exports assets. Godot 4 assembles assets at runtime based on functional blueprints; the authoritative standalone C++20 PlanetSim determines all physical, industrial and environmental behavior. **No language model or neural 3D model is shipped in the game runtime.**

The architecture preserves the existing Planet Game engineering sandbox: Materials, Processes, Components, typed Ports, Connections and hierarchical Blueprints; deterministic mass/energy accounting; knowledge-driven improvements; environmental coupling; Godot rendering with GDExtension/godot-cpp and headless C++20 simulation. PC remains the primary target, with Android adaptation later.

1. System architecture



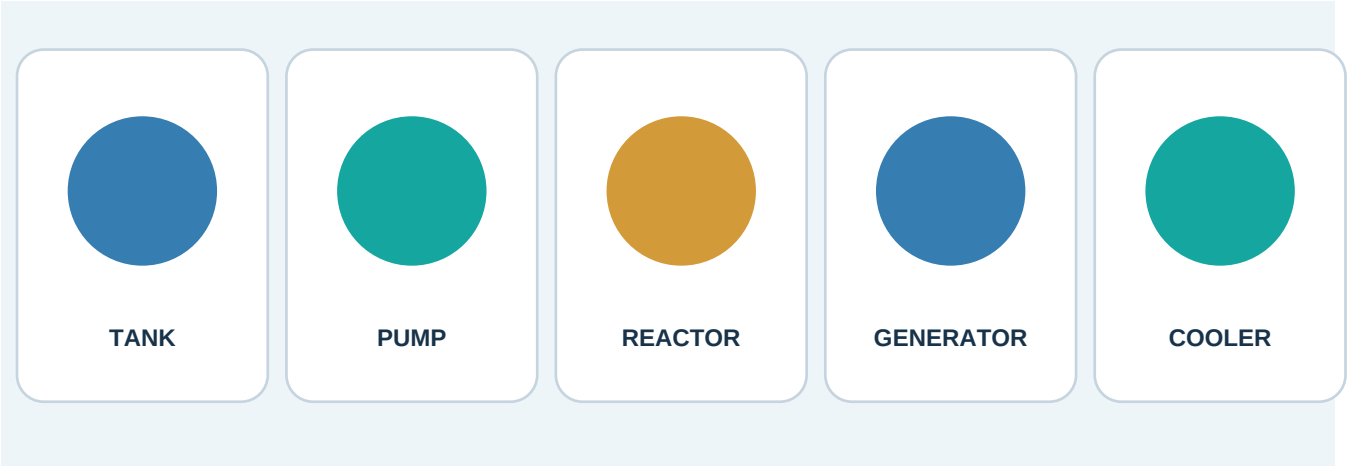
Stage	Responsibility	Output
GPT / Codex	Specify assets, write procedural generators and validation tests	Versioned JSON and Python
Concept-image stage	Supply appearance references for irregular assets	Approved image references
Neural 3D backend	Generate visual mesh and material candidates	Raw meshes and textures
Blender procedural backend	Create precise machinery, structural modules and connectors	Parametric geometry
Blender asset compiler	Normalize transforms, UVs, LODs, rigging, colliders and materials	Validated scene / GLB
Asset registry	Track semantic metadata, origin and dependencies	GLB + JSON + provenance
Godot 4	Assemble, animate, visualize, route pipes/cables	Runtime scenes
PlanetSim C++20	Authoritative processes, flows, resources, environmental effects	Deterministic state snapshots

Separation rules

1. The simulation must build and run without Godot, Blender, or any AI tools.
2. Mesh appearance must never define physical laws or override simulation results.
3. Functional ports and process definitions have stable identifiers shared by simulation and visual bindings.
4. Asset generation is offline; runtime assembly is deterministic and bounded by validated rules.
5. All generated content is reviewable, reproducible and license-traceable.

2. Asset categories and production strategies

Category	Preferred technique	Reason
Industrial machinery, factories	Blender Python / Geometry Nodes	Accurate ports, pivots, dimensions and modularity
Rocks, vegetation, decorative objects	Image-to-3D neural backend	Irregular organic forms
Roads, pipes, wires, conveyors	Godot procedural runtime assembly	Dynamic connections and junction routing
Vehicles and robots	Hybrid neural visual shell + procedural mechanical parts	Articulated movement and predictable physics
Terrain and planetary surfaces	Existing PlanetSim/planet generator pipeline	Planet-scale deterministic coupling



Neural backend policy

Begin with TRELLIS.2 as a *research prototype*, not a hard dependency. A model adapter should expose input image, seed, quality profile, output mesh, materials and provenance. Evaluate license terms before commercial use. In particular, the TRELLIS.2 project page contains a research/noncommercial-use disclaimer; Hunyuan3D-2.1's published community license excludes the EU. Do not assume generated meshes are production-ready. Ensure fallback to entirely procedural generation.

3. Functional asset contract

The defining concept is that a game asset carries **geometry + functional anchors + animation controls + material slots + validated metadata**, while PlanetSim owns the actual physical behavior.

An electric generator might expose mechanical input, electrical output and coolant input. Blender contains PORT_MECH_IN, PORT_ELEC_OUT, PORT_COOLANT_IN transforms and ANIM_ROTOR; an accompanying manifest references those nodes. The simulation determines power, temperature, wear and efficiency. Godot maps those values to rotor rotation, gauges, emission effects and heat tinting.

Example proposed manifest:

```
{
  "id": "generator_electric_v1",
  "type": "machine",
  "geometry": "generator.glb",
  "dimensions_m": [2.0, 1.5, 1.8],
  "components": ["housing", "rotor", "stator", "cooling_system"],
  "ports": [
    {"id": "mechanical_input", "type": "mechanical_power", "node": "PORT_MECH_IN"},
    {"id": "electric_output", "type": "electrical_power", "node": "PORT_ELEC_OUT"},
    {"id": "coolant_input", "type": "fluid", "node": "PORT_COOLANT_IN"}
  ],
  "animations": {"rotor_speed": "ANIM_ROTOR", "temperature": "MAT_HEAT"}
}
```

This is a **visual asset manifest**, not a replacement for PlanetSim's existing process schema. A binding/validation layer resolves semantic IDs, verifies port compatibility, and checks the transform names against imported GLB nodes.

4. Procedural factory assembly

An engineering blueprint describes processes and typed connections. A deterministic layout solver chooses footprints and placement, selects approved machine-module variants, routes fluid pipes / power cables / conveyor paths, then Godot instantiates the visual assembly. The solver checks port orientation, clearances, connectivity and feasible routing. It may report that a blueprint is impossible to lay out rather than inventing geometry.

Reusable functional modules

Module	Connections / state
Tank	Material inventory, fill level, temperature
Pump	Fluid input/output, electrical draw, throughput
Reactor	Process inputs/outputs, heat, pressure, emissions
Generator	Mechanical/fuel input, electrical output, thermal losses
Conveyor	Item flow, speed, utilization
Heat exchanger	Thermal and fluid interfaces
Research station	Knowledge production and technology unlocks
Pollution-control system	Emission treatment, waste and energy use

Three separate records are maintained: **functional blueprint** (semantics), **physical layout** (position/dimensions/routing) and **visual skin** (geometry/materials/animations). This allows fictionally invented equipment to be expressed through known component templates, without making arbitrary image-generation output physically authoritative.

5. Blender asset compiler

Compile process: import/generate -> normalize units/origin/transforms -> clean topology -> generate UV/materials -> build LOD and collision -> install functional anchors/pivots -> prepare animations -> validate -> export GLB, JSON and preview.

Validation includes: missing nodes/ports, dimensions, axis convention, attachment placement, duplicate semantic IDs, material/texture references, animation tracks, render budgets, collision geometry and Godot import smoke tests. Manifoldness checks depend on object category; open leaves or surfaces need not be watertight.

Blender .blend is the editable source; .glb is the interchange artifact. Core glTF supports meshes, textures, common PBR channels and animations. A stable Blender version should be pinned by the build environment.

Example CLI:

```
python tools/asset_pipeline/cli/build_asset.py \
    --spec assets/specifications/electric_generator.json \
    --quality high --target godot
```

6. Godot and PlanetSim bindings

Godot scenes contain foundation, modules, connection segments and visual effects. PlanetSim publishes a per-component state frame (e.g., flow, electric output, temperature, pressure, degradation, emissions). The visual binding maps each state variable to animation speed, shader parameters, indicators or particles. Render state is observational: it cannot feed authoritative state back by accident.

Runtime asset assembly is driven by deterministic seeds plus versioned blueprint and asset IDs; the same saved game should reconstruct the same arrangement. PlanetSim serialization contains no engine-owned pointers. Expensive rendering detail can be suppressed without changing production behavior.

7. Source layout

```
planet_game/
  tools/asset_pipeline/
    cli/ generators/{blender,neural}/
    processors/{geometry,materials,rigging,lod}/
    validators/ exporters/
  assets/
    specifications/ sources/{procedural,neural}/
    library/{machines,buildings,infrastructure,vehicles,vegetation,terrain}/
    generated/
  godot/
    procedural_assembly/ asset_registry/ simulation_visualization/
  tests/assets/
```

Maintain per-asset provenance: generator and commit versions, seeds, parameters, source images, licensing, model name, hashes, toolchain version and validation report. Keep large binaries in artifact storage or Git LFS while tracking definitions and scripts with Git.

8. Hardware strategy

Codex and Blender can run on the existing workstation. Neural 3D generation should use a queued, replaceable GPU worker with a local or remote implementation. TRELLIS.2 typically needs a high-VRAM NVIDIA device; the existing RTX 4070 Ti may not support full-quality runs. GPU availability must not block procedural asset builds or headless simulation tests.

9. Roadmap and acceptance criteria

Phase 1: Procedural foundation. Create schemas, compiler interface, Blender builder, validation and Godot import. Deliver a generator with at least three typed ports and an animated pivot.

Phase 2: Neural integration. Add concept-image-to-mesh adapter, processing, provenance and licensing review. Deliver one approved organic decorative asset imported by the same pipeline.

Phase 3: Factory assembly. Connect the engineering graph, layout solver, port routing and state-driven visuals. Deliver a five-module tank/pump/reactor/generator/cooler chain with deterministic save/load.

Phase 4: Production scaling. Add batch generation, searchable previews, human approval, cache, LOD policy and regression tests. Deliver repeatable asset builds from source specifications.

First end-to-end proof of concept

Generate Tank, Pump, Reactor, Generator and Cooler; connect them using their typed ports; produce an automatic footprint and routed pipelines; simulate an operating state in PlanetSim; animate flow, rotor speed, temperature and emissions; modify one blueprint parameter and verify a deterministic re-assembly. Every output passes validation and a Godot import test.

10. Design decisions

• Three modules: AssetForge.Generate (offline procedural/neural creation), AssetForge.Compile (Blender processing, export and validation), AssetForge.Assemble (Godot deterministic layout and binding).

• **No GPT or neural 3D inference in the shipped runtime.**

• Gameplay physics, chemistry and economy remain exclusively in PlanetSim.

• Neural meshes are candidates for human approval, not semantically reliable machine definitions.

• Use typed port transforms and stable identifiers as the integration contract.

• Introduce a reproducible and license-aware asset registry from day one.

References (checked October 2026)

• Microsoft TRELLIS.2 project: https://microsoft.github.io/TRELLIS.2/

• TRELLIS repository: https://github.com/microsoft/TRELLIS

• Tencent Hunyuan3D-2.1 license: https://github.com/Tencent-Hunyuan/Hunyuan3D-2.1/blob/main/LICENSE

• Godot 3D scene formats: https://docs.godotengine.org/en/stable/tutorials/assets_pipeline/importing_3d_scenes/available_formats.html

• Blender glTF exporter: https://docs.blender.org/manual/en/latest/addons/import_export/scene_gltf2.html

The vector illustrations in this document are original schematic concepts, not generated production 3D meshes.